The Journal of Open Source Software

pyopmspe11: A Python framework using OPM Flow for the SPE11 benchmark project

David Landa-Marbán [1¶] **and Tor H. Sandve** [1]

1 NORCE Norwegian Research Centre AS, Bergen, Norway ¶ Corresponding author

DOI: 10.xxxxxx/draft

Software

- Review
- Repository
- Archive

Editor:

Submitted: 28 June 2024
Published: unpublished

License
Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License (CC BY 4.0).

DRAFT

Summary

The imperative to achieve climate change goals and the increasing worldwide demand for energy have made carbon capture and storage (CCS) technology more relevant today. Since utilizing computational models is essential for planning large-scale CCS projects, then it is crucial to benchmark simulation tools to enhance confidence in the results. Inspired by a recent validation study for laboratory-scale CO_2 storage (Flemisch et al., 2024), a new comparative solution project (CSP) was launched to simulate both lab- and field-scale CO_2 storage (Nordbotten et al., 2024). As part of the Early Access team, the authors have developed and made open the `pyopmspe11` tool which facilitates reproducible solutions to the benchmark. This tool serves as a common starting point for developing and testing new simulation technology. It is expected that its impact will extend far beyond the initial benchmark study.

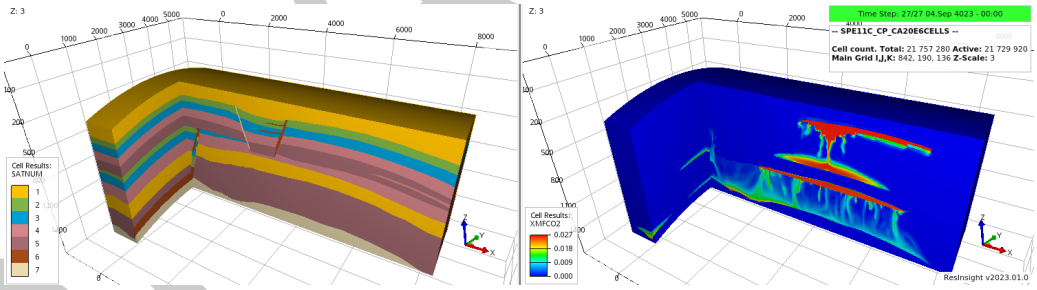

Figure 1: Generated model by the configuration file `spe11c_cp_ca20e6cells.txt` in the examples folder.

Statement of need

Geological carbon storage (GCS) applications benefit from both commercial and open-source simulators. However, using open-source simulators often involves data preprocessing and postprocessing, which can be challenging for those without advanced programming knowledge. Additionally, setting up and running simulations requires engineering expertise. To bridge this gap, developers can simplify the setup of numerical studies by using user-friendly approaches, such as configuration files. This not only ensures result reproducibility but also facilitates flexible testing of different simulator parameters and allows for easy extension to further studies.

`pyopmspe11` is a simplified and flexible Python tool for the three cases in the SPE Comparative Solution Project via configuration files. `pyopmspe11` relies on the OPM Flow numerical simulator (Rasmussen et al., 2021), where the implementation of the CO_2 model can be found in Sandve et al. (2021). The primary contribution of `pyopmspe11` lies in its data preprocessing and postprocessing capabilities. It offers flexibility in generating various types of grids, including Cartesian, tensor, and corner-point grids. These grids adhere to standard industry formats,

making them compatible not only with OPM Flow but also with other simulators. Additionally, pyopmspe11 supports different resolutions, having been tested to generate approximately 160 million cells. In the context of data postprocessing, pyopmspe11 not only generates the necessary reporting data as specified by the benchmark, but it also produces .png figures for rapid inspection of individual simulations and for making comparisons between different runs (e.g., to assess sensitivities). The postprocessing methods efficiently map non-overlapping cell values (both intensive and extensive quantities) between the simulation grid and the reporting grid.

Outlook

pyopmspe11 is designed for use by researchers, engineers, and students. During the preliminary intercomparison workshops for the benchmark study, the authors received positive feedback about the framework. Some participant groups have utilized pyopmspe11, particularly for grid generation. Additionally, the authors have been contacted to provide specific support on how to use the tool for setting up further studies, such as optimizing injection strategies. Looking ahead, the plan for pyopmspe11's future development includes extending its functionality to incorporate additional physical models (e.g., hydrogen storage, salt precipitation, biofilm effects).

Acknowledgements

The authors acknowledge funding from the Center for Sustainable Subsurface Resources (CSSR), grant nr. 331841, supported by the Research Council of Norway, research partners NORCE Norwegian Research Centre and the University of Bergen, and user partners Equinor ASA, Wintershall Dea Norge AS, Sumitomo Corporation, Earth Science Analytics, GCE Ocean Technology, and SLB Scandinavia. The authors also acknowledge funding from the HPC Simulation Software for the Gigatonne Storage Challenge project, grant nr. 622059, supported by Equinor ASA and CLIMIT DEMO/Gassnova.

References

Flemisch, B., Nordbotten, J. M., Fernø, M., Juanes, R., Both, J. W., Class, H., Delshad, M., Doster, F., Ennis-King, J., Franc, J., Geiger, S., Gläser, D., Green, C., Gunning, J., Hajibeygi, H., Jackson, S. J., Jammoul, M., Karra, S., Li, J., … Zhang, Z. (2024). The FluidFlower validation benchmark study for the storage of CO_2. *Transport in Porous Media*, *151*(5), 865–912. https://doi.org/10.1007/s11242-023-01977-7

Nordbotten, J. M., Ferno, M. A., Flemisch, B., Kovscek, A. R., & Lie, K.-A. (2024). The 11th Society of Petroleum Engineers Comparative Solution Project: Problem Definition. *SPE Journal*, *29*(05), 2507–2524. https://doi.org/10.2118/218015-PA

Rasmussen, A. F., Sandve, T. H., Bao, K., Lauser, A., Hove, J., Skaflestad, B., Klöfkorn, R., Blatt, M., Rustad, A. B., Sævareid, O., Lie, K.-A., & Thune, A. (2021). The open porous media flow reservoir simulator. *Computers & Mathematics with Applications*, *81*, 159–185. https://doi.org/10.1016/j.camwa.2020.05.014

Sandve, T. H., Gasda, S. E., Rasmussen, A., & Rustad, A. B. (2021). Convective dissolution in field scale CO2 storage simulations using the OPM flow simulator. In N. A. Røkke & K. Knuutila (Eds.), *TCCS-11. CO2 capture, transport and storage. Trondheim 22nd–23rd june 2021. Short papers from the 11th international trondheim CCS conference* (pp. 113–119). SINTEF Academic Press.